Phase-dressed leading term with a uniform cutoff

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Abstract

Unit 204: the transport-only part of Astra #20's programme D, taken because it is what the paper's tubular fibres $(0,\varepsilon]^d$ actually need. The dilation $u = bx$ turns the phase-dressed integral on $(0,b]^d$ into $b^{\sum_i (h_i+1)}$ times the unit-box integral at parameter $Nb^{\sum_i k_i}$ with phase and amplitude composed with the dilation (`phaseBoxCutoff_eq`); since $(\log Nc/\log N)^{m-1} \to 1$, the leading coefficient acquires exactly $b^{\sum_i (h_i+1) - 2\lambda \sum_i k_i} = \prod_{i\notin J} b^{h_i+1-2k_i\lambda}$ (`phaseBoxCutoff_tendsto`), the b-dependence of the paper's eq. `a_minus_m_explicit`, while the face integral becomes $\int \eta(b\pi u) J_p(\xi(b\pi u)) \prod_{i\notin J} u_i^{h_i - pk_i}\, du$, i.e. the amplitude and phase are read on the face of the box $(0,b]^d$ (`headline_phase_leading_cutoff`). Review v9 (units 198–203) reported no statement-level defects; its should-fix (write $J_p^{(\beta)}$, never S, for `phaseMoment` in the documentation) is applied in this branch. Zero `sorry`/`axiom`; 9081 jobs.

1 The things formalised

```
noncomputable def phaseBoxCutoff (d) (h k) (b   N :  ) (   ) :   :=
   u in cutoffBox d b,   u * ((  i, u i ^ h i) * quadKernel   (  u) (N *   i, u i
     ^ k i))
theorem phaseBoxCutoff_eq (hb : 0 < b) : phaseBoxCutoff d h k b   N
    = b ^ (  i, (h i + 1)) *   x in unitBox d,   (b • x) * ((  i, x i ^ h i)
      * quadKernel   (  (b • x)) ((N * b ^ (  i, k i)) *   i, x i ^ k i))
theorem cutoff_power_eq' (hb) :
    b ^ ((  (h i + 1) :  ) - 2 l (  k i :  )) = b ^ (  (h i + 1)) * (b ^ (  k i)) ^
      (-(2 l))
theorem phaseBoxCutoff_tendsto (hk) (hb) (hl) (h ) (hmin) (hatt) (h ) (h ) :
    Tendsto (fun N => phaseBoxCutoff (d+1) h k b   N
      / (N ^ (-(2 l)) * log N ^ (multCount (ratioExp h k) l - 1))) atTop
     (  (b ^ ((  (h i + 1) :  ) - 2 l (  k i :  ))
        * phaseCoeff h k l   (fun x =>   (b • x)) (fun x =>   (b • x))))
theorem headline_phase_leading_cutoff … :
    Tendsto (fun N => (  u in cutoffBox (m+1) b,   u * ((  u^h)
      * exp (-(  N²   u^(2k)) +   (N   u^k)   u))) / (N^(-(2l)) * log N ^
      (multCount … - 1)))
     atTop (  (b ^ (…) / ((multCount … - 1)! *   i, if ratioExp h k i = l then k i
      else 1)
        *   u in unitBox (m+1), (  (b • faceProj h k l u) * phaseMoment   (2l) (
      (b • faceProj h k l u)))
          * residualWeight h k l u))
```

2 Mathematics

Pure transport: `setIntegral_comp_smul_of_pos` for the dilation of the box, the scaling $N \mapsto Nb^{\sum k_i}$ inside the kernel (the phase argument $s = Nu^k$ scales exactly, so the dressed kernel is unchanged in form), and the two limits $I(Nc)/((Nc)^{-p}(\log Nc)^{m-1}) \to C$ and $(\log Nc/\log N)^{m-1} \to 1$ multiplied together, as in unit 183. Nothing new is estimated. Fixed *rectangular* cutoffs $\prod(0, a_i]$ would need the coordinatewise dilation (a diagonal linear map) and give $\prod_i a_i^{h_i+1-pk_i}$; they are not used by the paper, whose fibres are cubes, so they are left out.

3 Review v9 disposition

Pass at statement level for units 198–203 (constants, normalisations, tail lemmas, mirror prose). Should-fix applied: documentation writes $J_p^{(\beta)}(a)$ (or `phaseMoment`) for $\int_0^\infty s^{p-1}e^{-\beta s^2+\beta as}\,ds$, never $S_{p/2}$, which in the paper is twice this. Cosmetic applied: a naming note that the Lean parameter `m + 1` is the dimension while the prose $m = |J|$ is `multCount`; the Taylor prefactor $(\beta N)^j/j!$ made explicit in the unit-200 docstring; the swapped headline's wording. Mirror standing assumptions: added a pointer to the hypotheses of `eq:normal_moment_amplitude_lean` in the phase paragraph.

4 Lean notes

- `b • unitBox d` needs `open scoped Pointwise` (again), else "failed to synthesize instance".
- `continuous_const.smul continuous_id` does not elaborate against the expected `Continuous fun x => b • x` (it produces `(fun x => ?c) • id`); use `continuous_const_smul b`.
- Reusing `tendsto_log_scale_pow` and the `filter_upwards [eventually_gt_atTop (max 1 (1/c))]` template from `CutoffScaling.lean` made the asymptotic transport a copy with `2l` in place of `l` and $c = b^{\sum k_i}$ in place of $b^{2\sum k_i}$.

5 Status

Programme A (deterministic core) and the paper-relevant part of D are complete. Next: mirror sentence for the cutoff, consolidated report v2 (units 165, 170–204), then stop or await direction (Astra #21: do not start C).